Fiscal Year:	FY 2021	**Task Last Updated:**	FY 06/16/2021
PI Name:	Weber, Richard Ph.D.		
Project Title:	The Origin of Fragility in High-Temperature Oxide Liquids		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rweber@matsdev.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	847-612-8597
Organization Name:	Materials Development Inc.		
PI Address 1:	3090 N Daniels Ct		
PI Address 2:			
PI Web Page:			
City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Japanese Space Agency (JAXA) AO for Electrostatic Levitation Furnace (ELF) experiments on ISS
Start Date:	08/15/2019	**End Date:**	09/30/2022
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2022 per M. Sansoucie/MSFC (Ed., 1/28/21) June 2019: Note project had preliminary work performed Feb-August 2019		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1288 ; 80NSSC19K0518		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed project will fund the United States (US) portion of an international Japan Aerospace Exploration Agency (JAXA) project [1] to investigate the properties and structure of molten and supercooled metal oxides using the Electrostatic Levitation Furnace (ELF) on the International Space Station (ISS) and ground-based synchrotron x-ray and neutron studies. Under the proposed project, NASA investigators will implement experiments using the ISS ELF to provide some of the first results from the instrument for US investigators. Flight experiments on molten barium oxide-titanium dioxide pseudo-binary compositions will be performed during the first and second years of the project. Samples will be launched on H-II Transfer Vehicle-8 (HTV-8) in mid-2019 and a second set of samples is planned for launch on HTV-9 in 2020. The measured thermophysical properties (density, viscosity, and surface tension) will be correlated with atomic structure obtained from synchrotron x-ray and neutron methods and by using structure-property models in collaboration with the Japanese investigators. The third year will be devoted to analysis of flight data, completing characterization of samples, and publishing the results. Work will be published in a timely way consistent with the goal of providing data to the broader scientific community. The project team brings together the skills and experience needed to implement a complex research program. Reference: [1] S. Kohara, et al., “The origin of fragility in high-temperature oxide liquids - towards fabrication of novel non-equilibrium oxides,” JAXA flight project funded in 2017.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Developing an advanced understanding of the properties of liquids that can be transformed into glasses is a high value enterprise that benefits the $300B global glass materials industry. Specific advances expected from this research will impact development of functional glasses by providing high accuracy thermophysical property data that can be used to optimize models used to design materials from first principles.
Task Progress:	A set of sample compositions was prepared as 2.0 ± 0.1 mm diameter spheres. Samples were made from mixtures of high purity metal oxide powders that were weighed, blended in a ball mill, and fused using a laser hearth. Chunks of material were formed into spheroids suitable for levitation testing. Samples were fully melted and this typically resulted in dense and robust products that were not friable. Samples were sent to JAXA for ground-based testing that included the following: Vibration to evaluate potential fracture or powdering during launch operations; Exposure to humidity to evaluate potential degradation during storage ; X-ray examination (radiography) to detect internal voids or large cracks; Laser beam heating using the 980 nm wavelength similar to the ELF to verify heating. Flight experiments are planned for August 2021. JAXA has prepared a timeline for the work and MDI (Materials Development Inc.) is planning precise details of the measurements to be made. The surface tension of the molten samples was estimated by reviewing literature for similar compositions. The estimate was used to calculate the expected resonant frequency for mode 2 oscillation of the drop. The oscillation frequency was also calculated using surface tension values 100 mN/m larger and smaller than the estimate. This approach provides 30-40 Hz range of frequencies over which the sample will be excited to determine the resonant frequency of a liquid drop. Once this is done for each composition, the plan will be refined. Work is in progress to evaluate different algorithms to convert the raw video data from ELF into .avi format that will be used for analysis of sample size to extract density as a function of temperature. Ground based measurements focused on neutron and x-ray diffraction measurements of glass samples made by levitation. A neutron diffraction isotope substitution method was used at the NOMAD (Nanoscale-Ordered Materials Diffractometer) beamline at the Spallation Neutron Source (SNS) at Oak Ridge National Laboratory in order to determine the partial structure factors for neodymium and titanium in a neodymium titanate composition glass. Complementary x-ray measurements were made at the Advanced Photon Source at Argonne National Laboratory. Two papers are in preparation. Abstracts have been submitted to both the ISS Research & Development conference that will be held virtually in August 2021 and the American Society for Gravitational & Space Research (ASGSR) annual meeting that will be held in person in Baltimore in early November 2021.
Bibliography Type:	Description: (Last Updated: 06/06/2026)
Articles in Peer-reviewed Journals	Benmore CJ, González GB, Alderman OLG, Wilke SK, Yarger JL, Leinenweber K, Weber JKR. "Hard x-ray methods for studying the structure of amorphous thin films and bulk glassy oxides." J Phys Condens Matter. 2021 May 12;33(19):194001. https://doi.org/10.1088/1361-648X/abe352 ; PMID: 33540391 , May-2021